

Matrix Series and the Matrix Exponential

Swapnil Ray

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1 Norms on Matrix Spaces

Throughout, $M_n(\mathbb{C})$ denotes the algebra of $n \times n$ complex matrices, and $M_n(\mathbb{R})$ the real analogue. We identify $M_n(\mathbb{C})$ with $\mathbb{C}^{n^2}$ when topological or metric notions are needed.

Definition 1.1 (Matrix norm). A *matrix norm* on $M_n(\mathbb{C})$ is a function $\|\cdot\| : M_n(\mathbb{C}) \rightarrow [0, \infty)$ satisfying, for all $A, B \in M_n(\mathbb{C})$ and $\lambda \in \mathbb{C}$:

- (i) $\|A\| = 0 \iff A = 0$;
- (ii) $\|\lambda A\| = |\lambda| \|A\|$;
- (iii) $\|A + B\| \leq \|A\| + \|B\|$;
- (iv) $\|AB\| \leq \|A\| \|B\|$ (submultiplicativity).

Example 1.2. The following are submultiplicative matrix norms on $M_n(\mathbb{C})$:

- (a) the *max column sum norm*

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}|;$$

- (b) the *max row sum norm*

$$\|A\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|;$$

- (c) the *spectral norm*

$$\|A\|_2 = \sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^*A)}.$$

Verification of submultiplicativity in (a). Let $A = (a_{ij})$, $B = (b_{ij})$, and $C = AB = (c_{ij})$. Then

$$|c_{ij}| \leq \sum_{k=1}^n |a_{ik}| |b_{kj}|.$$

Fix j and sum over i :

$$\sum_{i=1}^n |c_{ij}| \leq \sum_{i=1}^n \sum_{k=1}^n |a_{ik}| |b_{kj}| = \sum_{k=1}^n |b_{kj}| \sum_{i=1}^n |a_{ik}| \leq \|A\|_1 \sum_{k=1}^n |b_{kj}| \leq \|A\|_1 \|B\|_1.$$

Taking the maximum over j gives $\|AB\|_1 \leq \|A\|_1 \|B\|_1$. The other cases are analogous (for $\|\cdot\|_2$ use $\|AB\|_2 \leq \|A\|_2 \|B\|_2$, a standard fact from singular value theory). $\square$

Remark 1.3. All norms on the finite-dimensional vector space $M_n(\mathbb{C})$ are equivalent, so the choice of norm does not affect convergence statements. We shall freely use any submultiplicative norm.

Lemma 1.4 (Powers and products). *For any submultiplicative norm and any $A, B \in M_n(\mathbb{C})$:*

- (i) $\|A^k\| \leq \|A\|^k$ for all $k \geq 1$;
- (ii) $\|A^k B\| \leq \|A\|^k \|B\|$ and $\|BA^k\| \leq \|B\| \|A\|^k$.

Proof. (i) By induction on k . The case $k = 1$ is $\|A\| \leq \|A\|$. For $k \geq 1$,

$$\|A^{k+1}\| = \|A^k A\| \leq \|A^k\| \|A\| \leq \|A\|^{k+1}.$$

(ii) Immediate from submultiplicativity. □

Remark 1.5. For a submultiplicative norm one has $\|I\| = \|I^2\| \leq \|I\|^2$, so $\|I\| \geq 1$. If $\|\cdot\|$ is a submultiplicative norm with $\|I\| \neq 1$, the norm $\|A\|' = \sup_{\|B\| \leq 1} \|AB\|$ satisfies $\|I\|' = 1$ and is still submultiplicative. Hence we may assume $\|I\| = 1$ without loss of generality.

2 Convergence of Matrix Series

Definition 2.1 (Convergence of a matrix series). Let $(A_k)_{k \geq 0}$ be a sequence in $M_n(\mathbb{C})$. The series $\sum_{k=0}^{\infty} A_k$ *converges* if the sequence of partial sums $S_N = \sum_{k=0}^N A_k$ converges in $M_n(\mathbb{C})$ (equivalently, in $\mathbb{C}^{n^2}$). It *converges absolutely* if $\sum_{k=0}^{\infty} \|A_k\| < \infty$.

Theorem 2.2 (Absolute convergence implies convergence). *If $\sum_{k=0}^{\infty} \|A_k\| < \infty$, then $\sum_{k=0}^{\infty} A_k$ converges in $M_n(\mathbb{C})$, and*

$$\left\| \sum_{k=0}^{\infty} A_k \right\| \leq \sum_{k=0}^{\infty} \|A_k\|.$$

Proof. Since $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$ is complete, it suffices to show that (S_N) is Cauchy. For $N > M$,

$$\|S_N - S_M\| = \left\| \sum_{k=M+1}^N A_k \right\| \leq \sum_{k=M+1}^N \|A_k\|.$$

The right-hand side tends to 0 as $M, N \rightarrow \infty$ because $\sum \|A_k\|$ converges. Hence (S_N) is Cauchy and therefore converges to some $S \in M_n(\mathbb{C})$.

For the norm inequality, continuity of the norm gives

$$\|S\| = \left\| \lim_{N \rightarrow \infty} S_N \right\| = \lim_{N \rightarrow \infty} \|S_N\| \leq \lim_{N \rightarrow \infty} \sum_{k=0}^N \|A_k\| = \sum_{k=0}^{\infty} \|A_k\|. \quad \square$$

Theorem 2.3 (Power series in a matrix). *Let $f(z) = \sum_{k=0}^{\infty} c_k z^k$ be a scalar power series with radius of convergence $R \in (0, \infty]$. Then for every $A \in M_n(\mathbb{C})$ with $\|A\| < R$, the matrix series*

$$f(A) = \sum_{k=0}^{\infty} c_k A^k$$

converges absolutely (and hence converges). Moreover, the convergence is uniform on every closed ball $\{A : \|A\| \leq r\}$ with $r < R$.

Proof. Let $r < R$. For $\|A\| \leq r$, Lemma 1.4 gives

$$\|c_k A^k\| \leq |c_k| \|A\|^k \leq |c_k| r^k.$$

Since $r < R$, the scalar series $\sum_{k=0}^{\infty} |c_k| r^k$ converges. By the Weierstrass M -test, $\sum c_k A^k$ converges absolutely and uniformly on $\{\|A\| \leq r\}$. Taking $r \uparrow R$ gives absolute convergence whenever $\|A\| < R$. $\square$

Corollary 2.4. *Under the hypotheses of Theorem 2.3, the map $A \mapsto f(A)$ is continuous on $\{\|A\| < R\}$; indeed it is analytic there.*

Proof. Uniform convergence of a series of continuous (polynomial) functions on each closed ball $\{\|A\| \leq r\}$, $r < R$, implies continuity on each such ball, hence on the union $\{\|A\| < R\}$. Analyticity follows from the same argument applied to the power series expansion about any point A_0 with $\|A_0\| < R$. $\square$

3 The Matrix Exponential

Definition 3.1 (Matrix exponential). For $A \in M_n(\mathbb{C})$, define

$$\exp(A) = \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

Theorem 3.2 (Convergence of the exponential). *For every $A \in M_n(\mathbb{C})$, the series $\exp(A)$ converges absolutely. Moreover, for every $r > 0$ the convergence is uniform on $\{A : \|A\| \leq r\}$, and*

$$\|\exp(A)\| \leq e^{\|A\|}.$$

Proof. The scalar series $\sum_{k=0}^{\infty} z^k/k!$ has radius of convergence $R = \infty$. Apply Theorem 2.3 with $c_k = 1/k!$ and $R = \infty$. The estimate

$$\|\exp(A)\| \leq \sum_{k=0}^{\infty} \frac{\|A\|^k}{k!} = e^{\|A\|}$$

follows from Theorem 2.2. $\square$

4 Algebraic Properties of the Exponential

Proposition 4.1 (Commuting case). *If $A, B \in M_n(\mathbb{C})$ satisfy $AB = BA$, then*

$$\exp(A + B) = \exp(A) \exp(B).$$

Proof. Let $S_N(A) = \sum_{k=0}^N A^k/k!$. Because A and B commute, the binomial theorem gives

$$(A + B)^m = \sum_{k=0}^m \binom{m}{k} A^k B^{m-k}.$$

Hence

$$S_N(A)S_N(B) = \sum_{j=0}^N \sum_{k=0}^N \frac{A^j B^k}{j!k!} = \sum_{m=0}^{2N} \sum_{\substack{j+k=m \\ 0 \leq j, k \leq N}} \frac{A^j B^k}{j!k!}.$$

For $0 \leq m \leq N$, the constraint $j, k \leq N$ is automatic, and the inner sum equals

$$\sum_{j=0}^m \frac{A^j B^{m-j}}{j!(m-j)!} = \frac{1}{m!} \sum_{j=0}^m \binom{m}{j} A^j B^{m-j} = \frac{(A+B)^m}{m!}.$$

Therefore

$$S_N(A)S_N(B) - S_N(A+B) = \sum_{m=N+1}^{2N} \sum_{\substack{j+k=m \\ 0 \leq j, k \leq N}} \frac{A^j B^{m-k}}{j!k!}.$$

Taking norms and using $\|A^j B^{m-j}\| \leq \|A\|^j \|B\|^{m-j}$,

$$\|S_N(A)S_N(B) - S_N(A+B)\| \leq \sum_{m=N+1}^{2N} \frac{(\|A\| + \|B\|)^m}{m!} \leq \sum_{m=N+1}^{\infty} \frac{(\|A\| + \|B\|)^m}{m!} \rightarrow 0$$

as $N \rightarrow \infty$, since the exponential series converges. Hence $S_N(A)S_N(B) \rightarrow \exp(A+B)$. Since $S_N(A) \rightarrow \exp(A)$ and $S_N(B) \rightarrow \exp(B)$, continuity of matrix multiplication gives $\exp(A)\exp(B) = \exp(A+B)$. $\square$

Corollary 4.2. *For every $A \in M_n(\mathbb{C})$, $\exp(A)$ is invertible, with*

$$\exp(A)^{-1} = \exp(-A).$$

Proof. We first prove the identity $\exp(A)\exp(B) = \exp(A+B)$ when $AB = BA$ (Proposition 4.1 below). Applying it with $B = -A$ (which commutes with A trivially) gives

$$\exp(A)\exp(-A) = \exp(A-A) = \exp(0) = I.$$

Similarly $\exp(-A)\exp(A) = I$. Hence $\exp(-A) = \exp(A)^{-1}$. $\square$

Proposition 4.3 (Basic identities). *For all $A \in M_n(\mathbb{C})$:*

- (i) $\exp(0) = I$;
- (ii) $\exp((s+t)A) = \exp(sA)\exp(tA)$ for all $s, t \in \mathbb{C}$;
- (iii) $\exp(nA) = \exp(A)^n$ for every $n \in \mathbb{Z}$;
- (iv) $\exp(PAP^{-1}) = P\exp(A)P^{-1}$ for every invertible $P \in GL_n(\mathbb{C})$;
- (v) $\exp(A^*) = \exp(A)^*$;
- (vi) $\exp(\bar{A}) = \overline{\exp(A)}$.

Proof. (i) Only the $k = 0$ term survives: $\exp(0) = I$.

(ii) Since sA and tA commute, this is a special case of Proposition 4.1.

(iii) Iterate (ii) with $s = t = 1$ for $n \geq 1$, and use Corollary 4.2 for $n < 0$.

(iv) For every $k \geq 0$, $(PAP^{-1})^k = PA^kP^{-1}$. Hence for every partial sum S_N ,

$$\sum_{k=0}^N \frac{(PAP^{-1})^k}{k!} = P \left(\sum_{k=0}^N \frac{A^k}{k!} \right) P^{-1}.$$

Letting $N \rightarrow \infty$ and using continuity of $B \mapsto PBP^{-1}$ gives the result.

(v) $(A^*)^k = (A^k)^*$, and conjugation is continuous, so

$$\exp(A^*) = \sum_{k=0}^{\infty} \frac{(A^k)^*}{k!} = \left(\sum_{k=0}^{\infty} \frac{A^k}{k!} \right)^* = \exp(A)^*.$$

(vi) Identical argument with complex conjugation. □

Example 4.4 (Failure of $\exp(A + B) = \exp(A)\exp(B)$). Let

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then $A^2 = B^2 = 0$, so

$$\exp(A) = I + A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \exp(B) = I + B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Hence

$$\exp(A)\exp(B) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix},$$

whereas

$$\exp(B)\exp(A) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Thus $\exp(A)\exp(B) \neq \exp(B)\exp(A)$; in particular, since $\exp(A + B) = \exp(A)\exp(B)$ would force equality, the identity fails. Also note $AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, as expected.

Proposition 4.5 (Trace–determinant formula). *For every $A \in M_n(\mathbb{C})$,*

$$\det(\exp(A)) = e^{\operatorname{tr}(A)}.$$

Proof. Step 1: Reduction to upper triangular form. By the Schur decomposition (or the complex Jordan form) there is a unitary U with

$$A = UTU^*, \quad T = (t_{ij}) \text{ upper triangular}.$$

Then $\exp(A) = U\exp(T)U^*$ by Proposition 4.3(iv), so $\det(\exp(A)) = \det(\exp(T))$ and $\operatorname{tr}(A) = \operatorname{tr}(T)$. It suffices to prove the result for upper triangular T .

Step 2: The case of upper triangular T . As T^k is upper triangular for all $k \geq 1$, $\exp(T)$ is also upper triangular. Also note that if $t_{11}, \dots, t_{nn}$ are the diagonal entries of T , then the diagonal entries of $\exp(T)$ are

$$e^{t_{11}}, \dots, e^{t_{nn}}.$$

As $\exp(T)$ is upper triangular

$$\det(\exp(T)) = \prod_{i=1}^n e^{t_{ii}} = e^{\sum_i t_{ii}} = e^{\operatorname{tr}(T)}.$$

This proves the claim. □

Proposition 4.6 (Positive definiteness for Hermitian matrices). *If A is Hermitian ($A^* = A$), then $\exp(A)$ is Hermitian and positive definite. If A is skew-Hermitian ($A^* = -A$), then $\exp(A)$ is unitary.*

Proof. Hermitian case: by Proposition 4.3(v), $\exp(A)^* = \exp(A^*) = \exp(A)$, so $\exp(A)$ is Hermitian. By the spectral theorem, $A = U \operatorname{diag}(\lambda_1, \dots, \lambda_n) U^*$ with $\lambda_i \in \mathbb{R}$. Then

$$\exp(A) = U \operatorname{diag}(e^{\lambda_1}, \dots, e^{\lambda_n}) U^*,$$

which has positive eigenvalues $e^{\lambda_i} > 0$. Hence $\exp(A)$ is positive definite.

Skew-Hermitian case: $\exp(A)^* \exp(A) = \exp(A^*) \exp(A) = \exp(-A) \exp(A) = \exp(0) = I$, so $\exp(A)$ is unitary. $\square$

5 Analytic Properties of the Exponential

Proposition 5.1 (Derivative of the exponential along a line). *For every $A \in M_n(\mathbb{C})$ and every $t \in \mathbb{R}$,*

$$\frac{d}{dt} \exp(tA) = A \exp(tA) = \exp(tA) A.$$

Proof. Consider the difference quotient

$$\frac{\exp((t+h)A) - \exp(tA)}{h}.$$

Because tA and hA commute, Proposition 4.1 gives

$$\exp((t+h)A) = \exp(tA) \exp(hA).$$

Hence

$$\frac{\exp((t+h)A) - \exp(tA)}{h} = \exp(tA) \cdot \frac{\exp(hA) - I}{h}.$$

Now

$$\frac{\exp(hA) - I}{h} = A + \frac{hA^2}{2!} + \frac{h^2A^3}{3!} + \dots = A + h \sum_{k=2}^{\infty} \frac{h^{k-2}A^k}{k!}.$$

The series on the right converges absolutely and is bounded in norm by $\sum_{k=2}^{\infty} \frac{|h|^{k-2} \|A\|^k}{k!} = \|A\|^2 \sum_{m=0}^{\infty} \frac{|h|^m \|A\|^m}{(m+2)!} \leq \|A\|^2 e^{|h| \|A\|}$, which stays bounded as $h \rightarrow 0$. Hence the limit as $h \rightarrow 0$ is $\exp(tA)A$. The identity $A \exp(tA) = \exp(tA)A$ follows because A commutes with every partial sum of $\exp(tA)$ and multiplication is continuous. $\square$

Corollary 5.2 (Solution of linear ODEs). *For every $A \in M_n(\mathbb{C})$ and every $x_0 \in \mathbb{C}^n$, the unique solution of the initial value problem*

$$x'(t) = Ax(t), \quad x(0) = x_0$$

is $x(t) = \exp(tA)x_0$.

Proof. By Proposition 5.1,

$$\frac{d}{dt} \exp(tA)x_0 = A \exp(tA)x_0 = Ax(t),$$

so $x(t)$ solves the ODE and satisfies $x(0) = \exp(0)x_0 = x_0$. Uniqueness follows from the standard Picard–Lindelöf theorem, or directly by differentiating $e^{-tA}x(t)$:

$$\frac{d}{dt}(e^{-tA}x(t)) = -Ae^{-tA}x(t) + e^{-tA}Ax(t) = 0,$$

so $e^{-tA}x(t)$ is constant, equal to x_0 at $t = 0$, whence $x(t) = e^{tA}x_0$. $\square$

Proposition 5.3 (Local diffeomorphism). *The exponential map $\exp : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$ is a local diffeomorphism near 0. More precisely, there exist open neighborhoods U of 0 and V of I such that $\exp : U \rightarrow V$ is a diffeomorphism.*

Proof. We first compute the Fréchet derivative of $\exp$ at 0. For $H \in M_n(\mathbb{C})$,

$$\exp(H) - \exp(0) = \exp(H) - I = H + \frac{H^2}{2!} + \frac{H^3}{3!} + \cdots.$$

Hence

$$\|\exp(H) - I - H\| \leq \sum_{k=2}^{\infty} \frac{\|H\|^k}{k!} \leq \|H\|^2 e^{\|H\|} = o(\|H\|) \quad (\|H\| \rightarrow 0).$$

Therefore $D\exp(0) = \text{id}_{M_n(\mathbb{C})}$, which is invertible. By the inverse function theorem (applied to the smooth map $\exp$ between the finite-dimensional vector spaces $M_n(\mathbb{C})$ and $M_n(\mathbb{C})$), there exist open neighborhoods U of 0 and V of I such that $\exp : U \rightarrow V$ is a diffeomorphism. $\square$

Theorem 5.4 (Jacobian of the exponential). *For every $A, B \in M_n(\mathbb{C})$, the Fréchet derivative of $\exp$ at A in the direction B is*

$$D\exp_B(A) = \int_0^1 e^{sA} B e^{(1-s)A} ds.$$

Proof. Consider the smooth curve $\gamma(t) = \exp(A + tB)$. Differentiate the defining series term by term. For each $k \geq 1$,

$$\frac{d}{dt}(A + tB)^k = \sum_{j=0}^{k-1} (A + tB)^j B (A + tB)^{k-1-j}.$$

Hence

$$\gamma'(t) = \sum_{k=1}^{\infty} \frac{1}{k!} \sum_{j=0}^{k-1} (A + tB)^j B (A + tB)^{k-1-j}.$$

Setting $t = 0$ and using the uniform convergence of the differentiated series (which follows from the same M -test argument), we obtain

$$D\exp_B(A) = \gamma'(0) = \sum_{k=1}^{\infty} \frac{1}{k!} \sum_{j=0}^{k-1} A^j B A^{k-1-j}.$$

Now use the integral representation

$$\frac{1}{k!} \sum_{j=0}^{k-1} A^j B A^{k-1-j} = \frac{1}{(k-1)!} \int_0^1 e^{sA} B e^{(1-s)A} s^{k-2} ds \dots$$

More directly, note that for any $s \in [0, 1]$,

$$e^{sA} B e^{(1-s)A} = \sum_{p,q \geq 0} \frac{s^p (1-s)^q}{p! q!} A^p B A^q.$$

Integrating term by term over $s \in [0, 1]$ and using $\int_0^1 s^p (1-s)^q ds = \frac{p! q!}{(p+q+1)!}$,

$$\int_0^1 e^{sA} B e^{(1-s)A} ds = \sum_{p,q \geq 0} \frac{A^p B A^q}{(p+q+1)!}.$$

Interchange of summation and integration is justified by absolute convergence of the double series. The right-hand side groups according to $k = p + q + 1$:

$$= \sum_{k=1}^{\infty} \frac{1}{k!} \sum_{j=0}^{k-1} A^j B A^{k-1-j},$$

which equals $D \exp_A(B)$ by the computation above. □